

Yikang (Ethan) Yu

Software Engineering | AI and ML Systems | Robotics

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Profile

Applied AI and software engineer with experience teaching humanoid robots to play table tennis, building AI tutoring and retrieval systems, and designing tools that help users understand and evaluate agent work. Background spans reinforcement learning, language-model fine-tuning, backend infrastructure, multimodal data processing, and controlled evaluation of language models.

Education

University of Southern California

M.S. in Applied Data Science | Los Angeles, CA | Jan 2026 - Expected Dec 2027

University of California Berkeley

B.A. in Data Science and Economics | Berkeley, CA | Dec 2024

Robotics Experience

AgiBot Zhiyuan Robotics

Research Intern | Shanghai, China | Jun 2026 - Sep 2026

HOPE humanoid table-tennis system

- Co-developed and deployed a system that teaches AgiBot's A3 humanoid robot, with **31 degrees of freedom**, to track an incoming table-tennis ball and return it with human-like forehand and backhand strokes.
- Tuned **Proximal Policy Optimization (PPO)** hyperparameters and revised the training model, helping improve measured table-tennis accuracy from the **60% range to over 90% for both forehand and backhand strokes** in project evaluations.
- Trained the whole-body control policy across **4,096 parallel Isaac Lab environments**. The policy learns to translate strike targets and robot state into coordinated joint commands; a held-out simulation benchmark recorded **256/256 legal forehand returns**.
- Used **BeyondMimic** motion tracking and human swing references to teach the robot how to imitate table-tennis strokes. Worked on whole-body control, coordinating arm swings with locomotion (body movement) and balance.
- Worked within a **HITTER-based** system that connects ball perception, strike planning, and learned control: motion capture supplies the ball's position, the planner chooses where and when to strike, and the policy controls the robot's movement.
- Calibrated a ball-flight model using **1,200+ motion-capture trajectories** to support ball-trajectory prediction and strike planning.
- Reduced median online strike-planning latency from **451 ms to 9.8 ms**, allowing the system to update its strike plan in real time as the ball moves.
- Deployed a **50 Hz whole-body policy** on the physical robot. A separate real-robot test recorded **11 legal returns in 12 trials, with zero falls**.

AI Systems Experience

UC Berkeley VIVE Center

Research Engineer | Berkeley, CA | Jan 2025 - Dec 2025

Undergraduate Research Assistant | Berkeley, CA | Nov 2023 - Dec 2024

TAI open-source AI teaching platform

- Built AI tutoring infrastructure that turns course materials and student questions into lessons with source citations and voice narration. Deployed text and voice tutoring in **UC Berkeley CS61A and CS294**, serving **100+ students**.
- Built a two-stage **retrieval-augmented generation (RAG)** backend that first finds relevant course documents and then ranks passages within them. Indexed **10,000+ documents** and improved retrieval accuracy from **38% to 60%**.
- Implemented token-aware recursive text chunking with four levels of fallback, file-level and chunk-level embeddings, and query reformulation to retrieve relevant material at both document and passage level.
- Developed a unified ingestion pipeline supporting **7+ formats** and **1,000+ pages**, including PDF, video, notebooks, HTML, and Markdown. Converted inputs into a shared Markdown representation for downstream indexing.
- Built video transcription using **Whisper** and speaker diarization using **pyannote**, making spoken course content available to the retrieval system.
- Refactored the **FastAPI** backend into query, generation, memory, and audio services. Added local **vLLM** model serving and wrapped ingestion and indexing services with Docker, caching, and idempotent retries.
- Designed **MongoDB-backed memory** that separates short-term conversation state from persistent user context, supporting continuity across tutoring sessions.
- Built a lesson-generation pipeline that plans an outline, generates pages grounded in retrieved material, and produces synchronized text-to-speech narration.
- Used **asyncio** to overlap outline and page generation and stream results to the user. Added block-level citation tracking and JSON, static HTML, and interactive HTML output modes.

Hefei City Cloud Data Center Co Ltd

Data Development Intern | Hefei, China | May 2023 - Jul 2023

- Co-developed a **GLM-based enterprise agent** for customer-support workflows, connecting it to an automotive partner's production database so responses could use business records.
- Designed multi-turn prompt state management and used error logs to investigate problems in the agent's reasoning, retrieval grounding, and output reliability.
- Built a taxonomy for production records and performed data cleaning and schema alignment to support structured retrieval by the agent.

Projects and Research

Clinical Agent Training and Evaluation

Research Project | 2026 | In progress

- Built a research environment for a **language-model doctor** to read the available medical history, ask questions, request tests, interpret returned results, and decide when to diagnose or finish. Developed separate workflows for ICU-record replay and learning tool use from clinical conversations.
- Developed a **MIMIC-IV data pipeline** that organizes patient, admission, ICU, laboratory, and procedure records into time-ordered case episodes. Used patient-level train/validation/test splits and kept hidden diagnoses, future events, and reference actions out of the doctor's visible observations.
- Implemented an interactive **patient simulator** that reveals recorded test results only after the matching order and result-availability time. The environment tracks pending results, advances a simulation clock, and returns an explicit unsupported outcome when a requested action has no recorded feedback.
- Built a preprocessing pipeline for **2,198 PMC-derived conversations covering 12 tool types**, validating tool/result pairing and required arguments, grouping similar visible text before splitting, and creating per-decision training examples from the available conversation history.
- Fine-tuned **Aloe-Beta-8B with 4-bit QLoRA**, using **1,742 training conversations and 21,595 decision targets**. Completed **2,700 optimizer updates**, with rank-16 adapters, an 8,192-token context window, and an effective batch size of eight.
- Implemented **completion-only supervised fine-tuning (SFT)**: masked context and tool-result tokens from the loss, normalized each conversation's training weight, and reweighted finish examples to address repeated tool calls and failure to stop.
- On **128 fixed validation prefixes from 95 cases**, improved action-schema validity from **104/128 to 128/128** and tool-name matching from **10/102 to 65/102** against recorded reference actions, comparing the base model with the selected SFT checkpoint.
- Built **Group Relative Policy Optimization (GRPO)** training infrastructure for the diagnostic simulator, including grouped trajectory sampling, relative-advantage normalization, clipped policy updates, and reward/constraint diagnostics. Evaluated protocol completion separately from diagnosis and diagnostic-workup agreement.
- Implemented **Slurm experiment orchestration** on USC CARC, with dependent data-preparation, training, and evaluation jobs; vLLM model serving; and checkpoint recovery of adapter, optimizer, random-state, and step information.
- Developed **Weights & Biases evaluation dashboards** and failure analysis for tool selection, full-argument matching, stopping behavior, and unsupported replay branches. Distinguished single-step action learning from complete workflow execution; strict tool-and-argument matching remained **13/102** on the fixed validation set.

Additional Projects and Research

Plan Execute - Human-Agent Collaboration Platform

Personal Project | Python, FastAPI, Next.js, SQLite | 2025 - Present

- Designed and built a **human-agent collaboration platform** that helps users understand what agents did, inspect supporting evidence, and ask contextual questions about their results. Connected work outlines, deliverables, and expandable explanations in a Python/FastAPI and Next.js application.
- Engineered a **versioned context bridge** from agent execution to an independent Study workspace, preserving source snapshots, citations, questions, and return locations so users can examine a result and return to the original work context.
- Implemented **SQLite-backed conversation persistence** and session- and source-version-scoped reading and playback state; restored drafts and positions across reloads and mode changes while keeping background agent execution independent.
- Built source-linked explanations with on-demand background concepts, contextual follow-up questions, and a Study Later queue that retains the user's question and captured evidence. Kept explanatory sessions separate from work execution and source-editing permissions.
- Developed regression checks and executed **end-to-end browser tests against live model services**, covering citation navigation, stale-source rejection, offline retry, draft restoration, and Work-to-Study continuity. Preserved explicit failure states when source versions or locations could not be resolved.
- Implemented **WebSocket streaming** and a durable audio-delivery queue, eliminating the **10-20% audio-message loss observed in the original implementation**. Retained revision history and permission boundaries for review of agent work.

Cross Lingual Clinical Decision Bias in LLMs

USC AI for Health Research | Advisor Prof. Ruishan Liu | 2026

Manuscript in preparation

- Designed a controlled study of whether changing the language of a medical question changes a language model's recommendations. Evaluated **9,150 question-language pairs across six languages**, with controls for translation quality.
- Measured counseling consistency and medical-myth accuracy using Gemini 2.0 Flash, Python, SciPy, and statsmodels, with statistical significance testing.
- Found a **14.9-percentage-point counseling consistency gap** and a **20-percentage-point medical-myth accuracy gap** between English and Swahili; accuracy was **73.7% versus 53.7%**, with $p < 10^{-6}$ for the reported accuracy comparison.

Technical Skills

Languages: Python, C++, Java, SQL.

Robot learning and control: PPO, motion imitation, BeyondMimic, whole-body control, locomotion and balance, Isaac Lab, MuJoCo, ROS 2, motion capture, ONNX.

AI and evaluation: PyTorch, RAG, vector search, file and passage embeddings, BGE/Qwen embeddings, vLLM, multi-agent systems, Whisper, pyannote, text-to-speech, SciPy, statsmodels, BLEU.

Model training and experimentation: SFT, QLoRA, GRPO, Hugging Face Transformers, PEFT, bitsandbytes, Slurm, Weights & Biases, checkpoint recovery, tool-use evaluation.

Backend and data systems: FastAPI, REST APIs, asyncio, WebSocket, MongoDB, SQLite, JSON Schema, Docker, Git, Linux, CI/CD, caching, idempotent retries, logging.